

AusCarbonCalc: Technical Brief & Project Documentation

Phase 1 • Embodied Carbon Telemetry Framework

Core Developers & Researchers:

- **Faezeh Akbari Nikjeh:** LCA Methodology, NCC Compliance Research, Threshold Calibration.
- **Mohsen JavdanFard:** Systems Architecture, IronPython Data Pipelines, Visual Programming Logic.

1. Project Overview

As the global construction sector transitions towards Net-Zero targets, mitigating upfront embodied carbon has become a primary objective. AusCarbonCalc is an open-source, ultra-lightweight parametric compliance framework integrated directly within the Rhino/Grasshopper environment. The tool provides instantaneous visual telemetry on embodied carbon metrics, empowering architects and researchers to execute true generative design iterations during Phase 1 (conceptual massing).

2. Methodology & System Architecture

The computational framework is built upon a hybrid architecture utilizing Grasshopper for geometric data extraction and an IronPython back-end core for logic execution and compliance checking.

To ensure absolute transparency and facilitate academic peer review, the script bypasses conventional "clustering," leaving the semantic filtering pipeline fully exposed. This allows users to track the data flow from geometry ingestion to the final color-coded rendering.

3. Computational Logic & Thresholds

The core algorithm calculates the structural carbon footprint with execution latencies of under 100 milliseconds. The upfront embodied carbon is computed dynamically using the following mathematical model:

$$EC_u = \sum (V_n \times \rho_n \times CF_n)$$

The tool evaluates the calculated carbon against strict baselines calibrated to the **Australian National Construction Code (NCC 2.0.0)** and localized **AusLCI** databases.

- **Absolute Carbon Budget:** 10,000,000 kg CO₂-e
- **Standard Concrete CF:** 320,000 kg CO₂-e/m³
- **Structural Steel CF:** 800,000 kg CO₂-e/m³
- **Timber / CLT CF:** -400,000 kg CO₂-e/m³ (Biogenic Sink)

The system output is linked to the Rhino viewport's real-time rendering engine, providing intuitive Green (Compliant) or Red (Non-Compliant) alerts based on the mathematical thresholds.

4. Publication Status & Full Manuscript Access

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The complete research manuscript detailing the empirical validation, lifecycle assessment methodology, and comparative analysis with global EC⁺ benchmarks is currently **Under Peer Review** for publication in a high-impact scientific journal.

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